Task 7: Create Jenkinsfile Defining Pipeline Stages

Objective

To create a declarative Jenkinsfile defining automated Build, Test, and Deploy stages.

Jenkinsfile

The Jenkinsfile was created in the root directory of the project.

It defines the following pipeline stages:

BUILD
↓
TEST
↓
DEPLOY

Build Stage : The Build stage checks the Python application for syntax errors:

```
python3 -m py_compile app.py
```

Test Stage : The Test stage executes automated tests:

```
python3 -m pytest
```

The test execution successfully produced:

```
1 passed
```

Deploy Stage : The Deploy stage executes the deployment script:

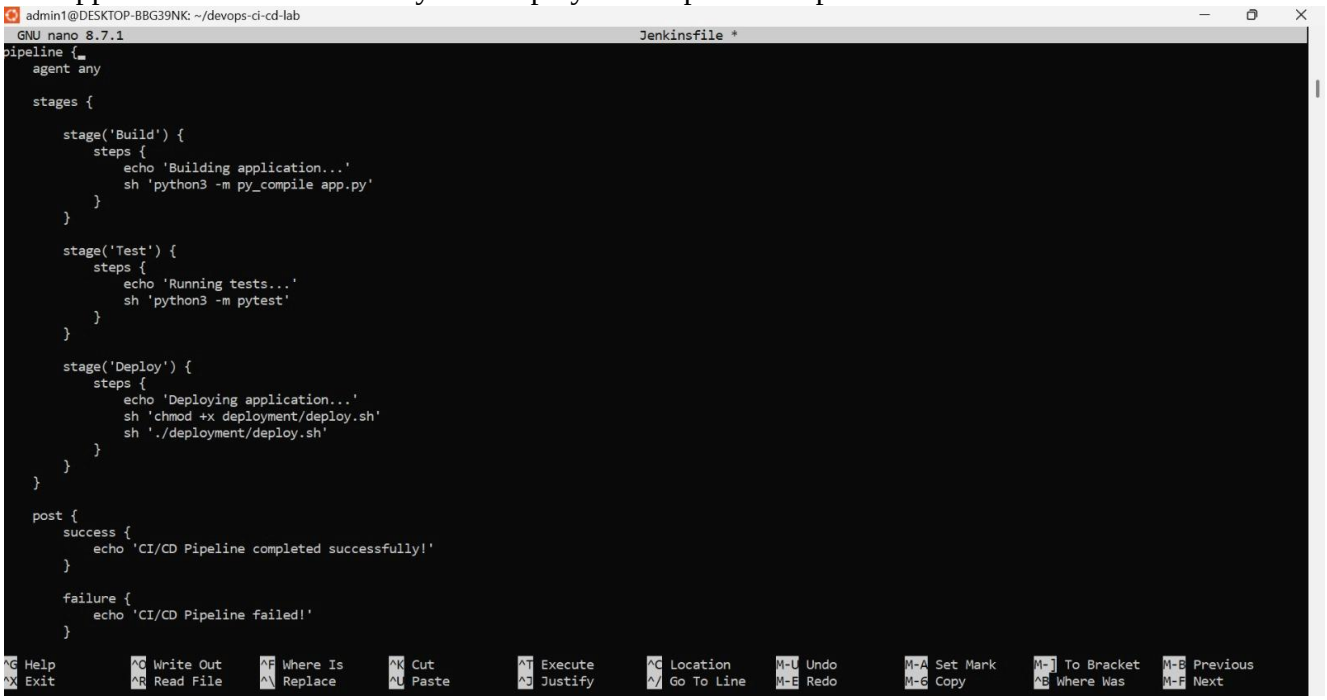
```
chmod +x deployment/deploy.sh
./deployment/deploy.sh
```

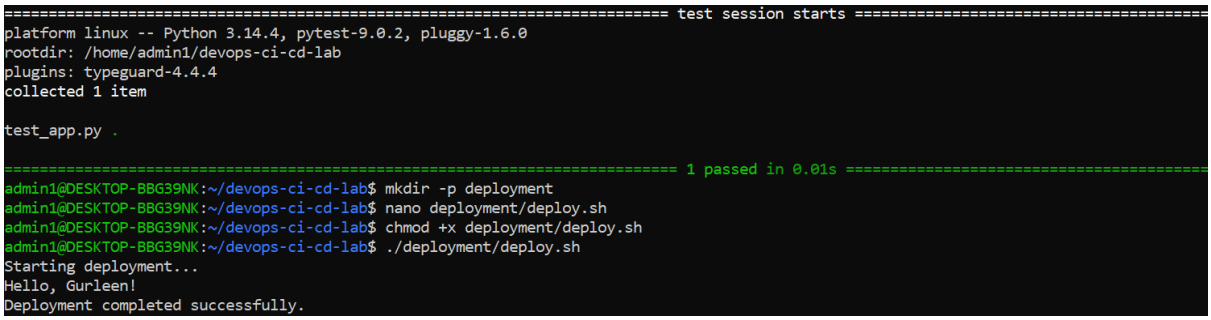
The deployment output was:

```
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
```

The complete DevOps workflow successfully integrated Git, GitHub, Jenkins, automated testing, and deployment. Source-code changes were managed using Git, stored on GitHub, automatically detected by Jenkins, and processed through Build, Test, and Deploy stages. The successful execution of the pipeline demonstrated a functional CI/CD workflow.

The application runs successfully and displays the expected output.

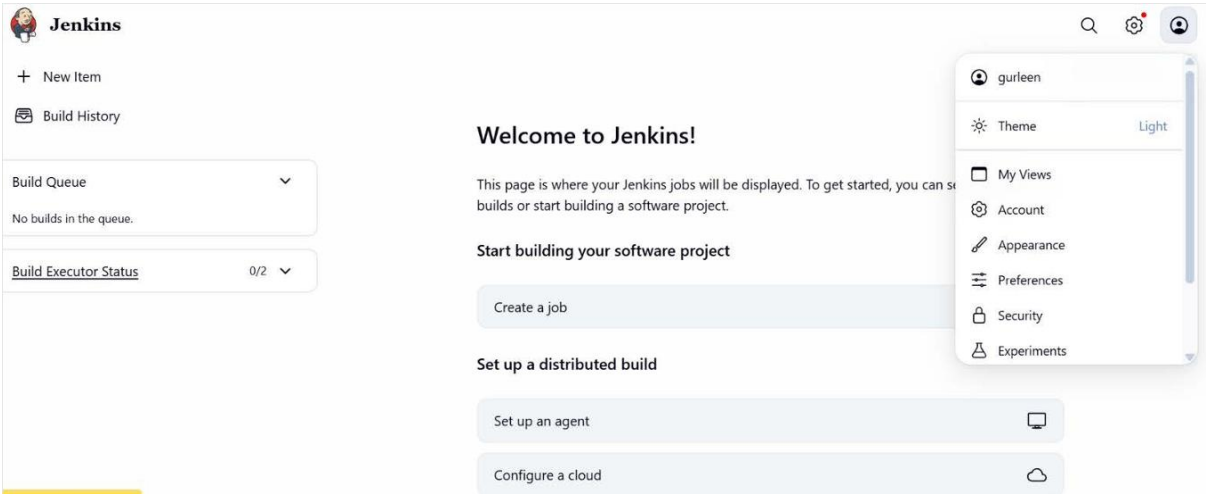


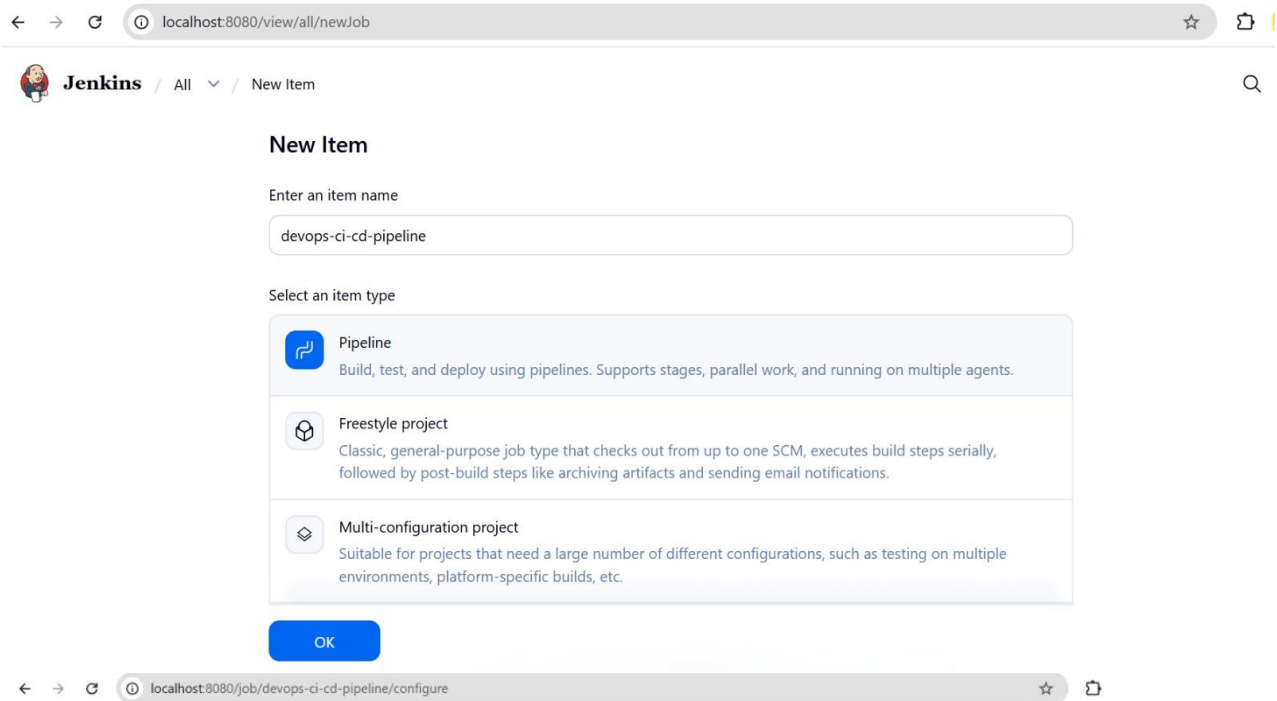


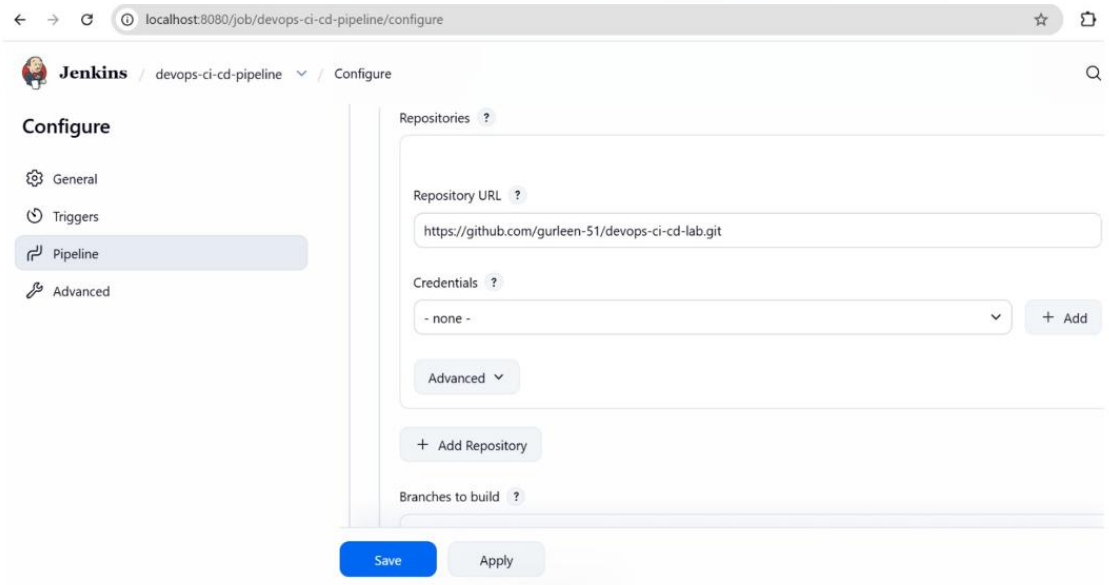
Jenkins automatically performed:

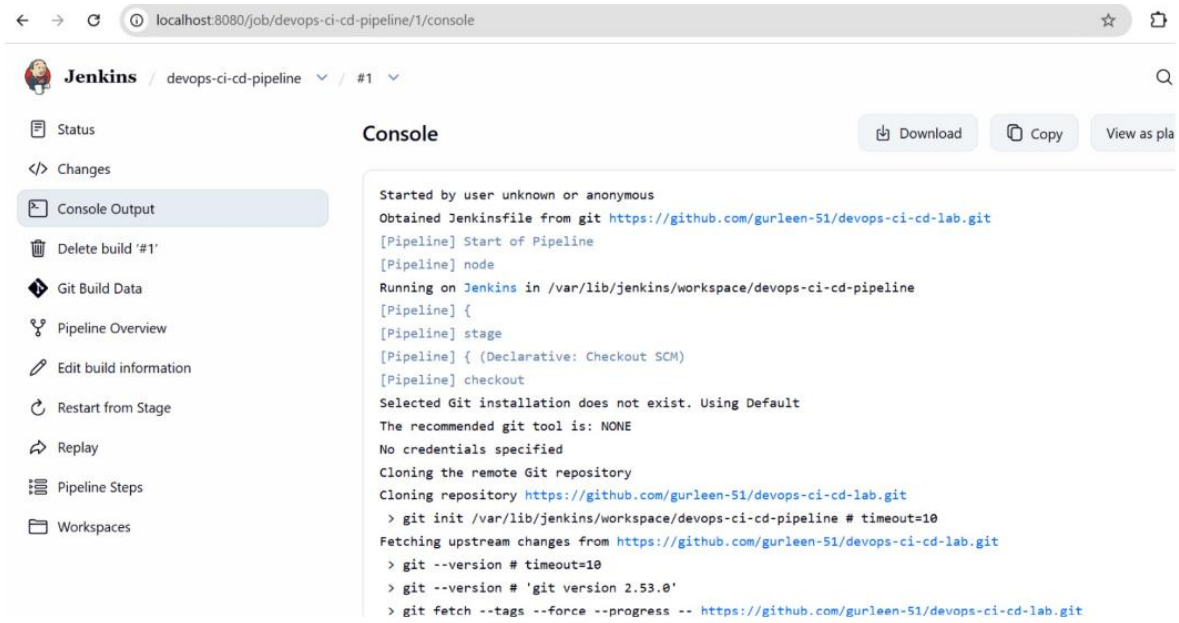
GitHub Change
↓
Jenkins Trigger
↓
Build
↓
Test
↓
Deploy

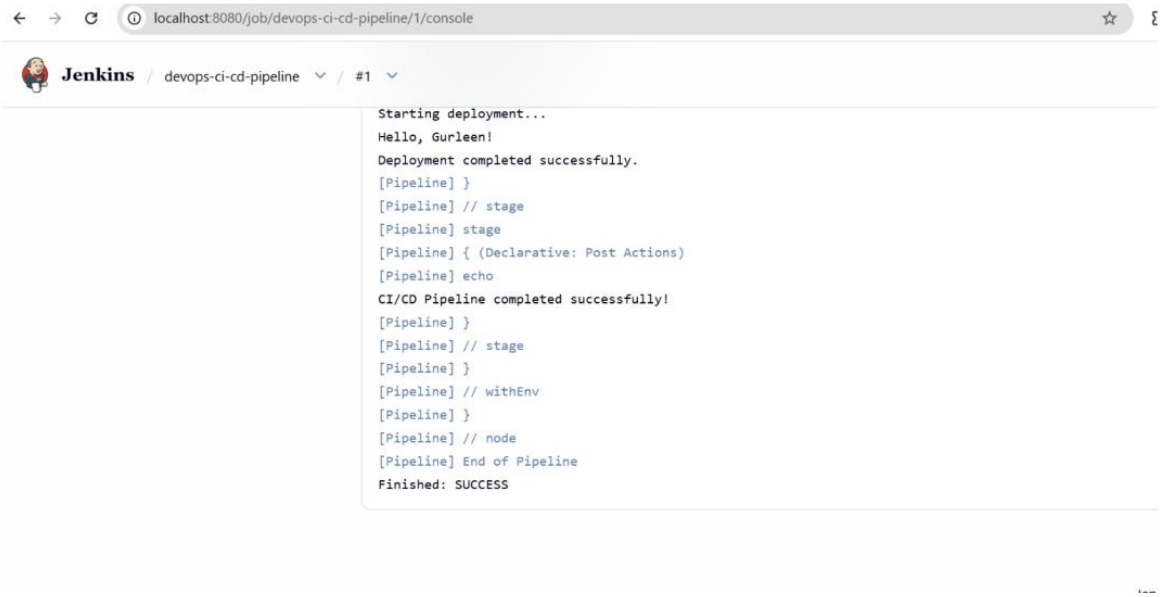
SUCCESS


Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status 0/2
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can s
builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
gurleen
Theme Light
My Views
Account
Appearance
Preferences
Security
Experiments


localhost:8080/view/all/newJob
Jenkins / All / New Item
New Item
Enter an item name
devops-ci-cd-pipeline
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
OK


localhost:8080/job/devops-ci-cd-pipeline/configure
Jenkins / devops-ci-cd-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Repositories
Repository URL
https://github.com/gurleen-51/devops-ci-cd-lab.git
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Save
Apply


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Status
Changes
Console Output
Delete build '#1'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as pla
Started by user unknown or anonymous
Obtained Jenkinsfile from git https://github.com/gurleen-51/devops-ci-cd-lab.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-ci-cd-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gurleen-51/devops-ci-cd-lab.git
> git init /var/lib/jenkins/workspace/devops-ci-cd-pipeline # timeout=10
Fetching upstream changes from https://github.com/gurleen-51/devops-ci-cd-lab.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
> git fetch --tags --force --progress -- https://github.com/gurleen-51/devops-ci-cd-lab.git


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS